Question			Answer	Marks	AOs	Guidance	
4	(a)		weight $\begin{pmatrix} 0 \\ -1.5g \end{pmatrix}$	B1	1.2	Allow seen or implied by correct answer	
			Equilibrium equation $\begin{pmatrix} 0 \\ -1.5g \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \end{pmatrix} + \mathbf{F}_2 = \mathbf{0}$	M1	1.1a	FT their weight. Allow sign errors	
			$\mathbf{F}_2 = \begin{pmatrix} -4 \\ 2+1.5g \end{pmatrix} = \begin{pmatrix} -4 \\ 16.7 \end{pmatrix}$ N	A1 [3]	1.1	must be vector allow **i-j** form	
4	(b)		Newton's second law $\begin{pmatrix} 0 \\ -1.5g \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \end{pmatrix} + \begin{pmatrix} 2 \\ 20 \end{pmatrix} = m\mathbf{a}$	M1	1.1a	Addition of vectors. Allow if weight missing but other two forces and acceleration seen in equation	Allow wrong weight only if given as a vector
			$\mathbf{a} = \frac{1}{1.5}\begin{pmatrix} 6 \\ 3.3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2.2 \end{pmatrix}$ m s^{-2}	A1 [2]	1.1 [2]	Must be vector; any correct form.	